HOSPITAL EMERGENCY ROOM ANALYTICS

Business Questions & Answers - 26 business questions answered end to end

Part 1 — Data Visualisation (Power BI)

Q1. How many total ER patients were seen, and what is the overall patient volume?

Answer: 9,216 total patient visits over the 19-month reporting window (April 2023 – October 2024), averaging roughly 485 visits per month.

Source: Power BI — Consolidated View, 'No. of Patients' KPI card.

Q2. What is the average patient wait time, and how does it trend month to month?

Answer: 35.3 minutes on average across all visits. The Monthly View trend line shows month-to-month fluctuation without a sustained upward or downward trend over the period.

Source: Power BI — Monthly/Consolidated View, 'Average Wait Time' KPI card and trend sparkline.

Q3. What is the average patient satisfaction score, and how does it trend?

Answer: 4.99 out of 10 among the visits with a recorded score. (Note: this average is based on only the ~27% of visits where a satisfaction score was captured — a data-quality caveat surfaced in the extended analysis, Q A1 below.)

Source: Power BI — 'Patient Satisfaction Score' KPI card; Key Takeaways page.

Q4. How many patients were referred to a specialty department, and which departments receive the most referrals?

Answer: 3,816 of 9,216 visits (41.4%) were referred to a specialty department. General Practice (1,840 referrals) and Orthopedics (995) are the top two by volume, followed by Physiotherapy (276), Cardiology (248), Neurology (193), Gastroenterology (178), and Renal (86).

Source: Power BI — 'No. of Patients Referred' KPI card and 'No of Patients by Department Referral' bar chart.

Q5. What proportion of patients are admitted versus treated and released?

Answer: 4,612 patients (50.0%) were admitted and 4,604 (50.0%) were treated and released — an almost even split.

Source: Power BI — 'Patient Admission Status' table.

Q6. What percentage of patients are seen within the 30-minute target, and what percentage miss it?

Answer: 40.7% of visits were seen within the 30-minute target; 59.3% missed it — meaning the ER misses its own timeliness target for a majority of patients.

Source: Power BI — '% of Patients Seen Within 30 Minutes' donut chart.

Q7. What is the age distribution of ER patients?

Answer: Volume is fairly even across age bands (1,056–1,200 visits per 10-year band), with the 30-39 group the single largest (1,200) and 20-29 close behind (1,188). No single age group dominates ER demand.

Source: Power BI — 'No of Patients by Age Group' bar chart.

Q8. What is the gender distribution of ER patients?

Answer: 51.0% male (4,705), 48.7% female (4,487), and 0.3% not confirmed (24) — close to an even split.

Source: Power BI — 'No, of Patients by Gender' donut chart.

Q9. What is the racial/ethnic composition of ER patients?

Answer: White (2,571) is the largest group, followed by African American (1,951), Two or More Races (1,557), Asian (1,060), Declined to Identify (1,030), Pacific Islander (549), and Native American/Alaska Native (498).

Source: Power BI — 'No of Patients by Patient Race' bar chart.

Q10. Which days of the week and hours of the day see the highest patient volume?

Answer: Saturday (1,377 visits) and Thursday (1,332) are the busiest days; Wednesday (1,260) is the quietest. By hour, volume peaks around 23:00 (436 visits), 07:00 (415), and 13:00 (410) — indicating late-night, early-morning, and early-afternoon surges rather than a single obvious peak.

Source: Power BI — 'No of Patients by Day & Hour' heatmap; Key Takeaways narrative.

Q11. Can an individual patient's visit details be looked up and filtered (ID, name, gender, age, admission date, race, wait time, department, admission status)?

Answer: Yes — the Patient Details page provides a fully filterable, sortable row-level grid of all 9,216 visits for drill-down and audit purposes.

Source: Power BI — Patient Details page.

Part 2 — Data Analysis (SQL, Python, Excel)

The next layer of analysis, going beyond what the dashboard shows — with the actual results obtained.

A. Data Quality & Survey Reliability

QA1. What share of ER visits has a recorded patient satisfaction score, and is the missing data random or systematically biased?

Answer: Only 27.3% of visits (2,517 of 9,216) have a satisfaction score — a 72.7% non-response rate. Formal tests show no significant bias by wait time (Welch's t-test, $p = 0.717$) or admission status (chi-square, $p = 0.260$); responders and non-responders look statistically similar on these fields. The response rate itself, not detectable bias, is the real caveat: the dashboard's 4.99/10 average should be reported with this context rather than as a clean, complete-sample figure.

Source: SQL (SQLite) + Python (Welch's t-test, chi-square)

QA2. Does the dataset contain structural data-quality issues — duplicate records, impossible ages/wait times, or inconsistent labels?

Answer: No. Automated checks found 0 duplicate Patient IDs, 0 age outliers (all ages between 0–110), and 0 wait-time outliers (all values between 0–300 minutes). The dataset passes a structural data-quality audit cleanly.

Source: Python (pandas validation rules) + Excel (audit sheet with live formulas)

B. Diagnostic & Statistical Analysis

QB1. Is the difference in average wait time across patient race groups statistically significant — i.e. is there evidence of inequitable service?

Answer: No. ANOVA (F-test, $p = 0.771$) and Kruskal-Wallis ($p = 0.775$) both fail to reject the null hypothesis. The visible spread in the dashboard's race chart is normal random variation, not evidence of race-based wait-time disparity. This is a genuinely reassuring, reportable finding for a health-equity review.

Source: Python (scipy.stats — one-way ANOVA, Kruskal-Wallis)

QB2. Is there a statistically significant relationship between wait time and satisfaction score, and is there a 'tipping point'?

Answer: No meaningful relationship. Pearson correlation $r \approx -0.02$ ($p \approx 0.29$) and Spearman r is similarly weak and non-significant. Contrary to the intuitive assumption built into the dashboard's framing, wait time does not meaningfully drive satisfaction in this data — something else (not captured here) is likely driving satisfaction scores.

Source: Python (Pearson & Spearman correlation, binned wait-time chart)

QB3. Which department referrals combine the longest wait times with the lowest satisfaction — where should resources be reallocated?

Answer: Neurology has the longest average wait (36.8 min), followed by Physiotherapy (36.6 min) and Gastroenterology (35.8 min). Renal has the shortest (34.7 min). The spread across departments is narrow (about 2 minutes end to end), so department-level wait time is not a major differentiator in this dataset — the bigger lever is the time-of-day staffing gap identified in Section D.

Source: SQL (SQLite) grouped aggregation + Excel department ranking table/chart

QB4. Is admission rate independent of age group and department referral?

Answer: Yes, independent on both. Chi-square tests return $p = 0.882$ (age group) and $p = 0.937$ (department referral) — both far above the 0.05 threshold. The near-50% admission rate holds steady across every age band and department; there's no segment that is a reliable early flag for admission likelihood.

Source: Python (chi-square test of independence)

C. Predictive Analytics & Machine Learning

QC1. Can we predict, at check-in, whether a patient will be admitted, using only age, gender, race, department referral, and arrival time?

Answer: No — both a Logistic Regression and a Random Forest classifier scored close to ROC-AUC 0.50 (equivalent to a coin flip). Admission is not predictable from check-in-time demographic and timing data alone in this dataset. The honest conclusion: a real admission-risk model would need clinical fields (triage acuity, chief complaint, vitals) that this dataset doesn't contain.

Source: Python (scikit-learn — Logistic Regression, Random Forest Classifier, ROC-AUC)

QC2. Can we predict a patient's expected wait time at check-in to give a realistic front-desk estimate?

Answer: No — both Linear Regression and Random Forest Regressor produced R^2 near zero (essentially no better than predicting the average for every patient), with a mean error of about 12.7 minutes. Wait time in this dataset behaves like noise with respect to age, department, and arrival timing; a useful version of this model would need a live queue-length feature.

Source: Python (scikit-learn — Linear Regression, Random Forest Regressor, MAE/RMSE/R^2)

QC3. Can patient satisfaction be predicted from operational variables, and which factor matters most?

Answer: Only weakly — a Random Forest classifier reached ROC-AUC ≈ 0.55, barely better than chance, and struggled to identify 'High' satisfaction cases. Consistent with the Q B2 result: wait time, admission outcome, department, and demographics don't explain satisfaction well in this data.

Source: Python (scikit-learn Random Forest classifier + feature importances)

QC4. Can patients be segmented into meaningful personas for targeted care pathways?

Answer: Yes — K-Means (k=4) with PCA visualization produced four clearly separated clusters, distinguished primarily by patient age and admission status (since wait time and demographics carry little separating signal on their own, per the findings above). This still gives a usable output for age-appropriate communication or care-pathway targeting, just not a wait-time optimization tool on this particular dataset.

Source: Python (scikit-learn — StandardScaler, K-Means, PCA)

D. Capacity Planning & Operational Strategy

QD1. What is the forecasted ER patient volume for the next 3 months, and should staffing budgets change?

Answer: The 30-day Holt-Winters forecast (weekly seasonality) projects an average of ≈16.1 patients/day, essentially flat against the historical average of ≈15.9/day. No volume growth is indicated, so no near-term staffing-budget increase is justified on volume alone — but the model confirms a real, reproducible day-of-week seasonality that should drive shift-level (not budget-level) staffing decisions.

Source: Python (statsmodels — Holt-Winters exponential smoothing, weekly seasonal_periods=7)

QD2. Where does demand most exceed service capacity, and where should the next shift's staff be placed?

Answer: The day×hour heatmap identifies the worst windows: Thursday 22:00-24:00 (70.7% of visits miss the 30-minute target), Saturday 22:00-24:00 (69.3%), and Monday 02:00-04:00 (68.5%). Late-night/overnight windows are consistently the weakest across nearly every day — the clearest, most actionable staffing signal in the entire analysis.

Source: SQL (SQLite) aggregation + Excel conditional-formatted heatmap

QD3. Is Case Manager (CM) assignment associated with better outcomes, and is coverage adequate?

Answer: Only about 5.2% of visits (480 of 9,216) have a CM assigned — far from a 50/50 split. CM-assigned visits show a modestly shorter average wait (34.4 vs. 35.3 minutes), but a Welch's t-test shows this is not statistically significant ($p = 0.211$). With coverage this thin, the data can't yet support a claim that the CM program measurably improves outcomes — a prospective pilot with expanded, more deliberately targeted CM assignment is the recommended next step.

Source: SQL (SQLite) grouped comparison + Python Welch's t-test

Overall takeaway: The dashboard in Part 1 tells us what happened and the analysis in Part 2 tests whether the visible patterns are statistically real, predicts forward, and translates findings into operational and financial recommendations — the layer that turns a reporting tool into a decision-support project.